

genres	
PK	genre_id (SERIAL)
	name (VARCHAR 100)

languages	
PK	language_id (SERIAL)
	name (VARCHAR 100)
	code (CHAR 2)

publishers	
PK	publisher_id (SERIAL)
	name (VARCHAR 200)
	country (VARCHAR 100)

book_conditions	
PK	condition_id (SERIAL)
	label (VARCHAR 50)
	description (TEXT)

authors	
PK	author_id (SERIAL)
	first_name (VARCHAR 100)
	last_name (VARCHAR 100)
	birth_year (SMALLINT)
	nationality (VARCHAR 100)

books	
PK	book_id (SERIAL)
	isbn (VARCHAR 20)
	title (VARCHAR 300)
	author (VARCHAR 200)
	year_published (SMALLINT)
	FK publisher_id genre_id language_id author_id

book_authors	
FK	author_id
FK	book_id

wishlist [M:N bridge]	
PK	wishlist_id (SERIAL)
FK	user_id
FK	book_id
	added_at (TIMESTAMP)

messages [TERNARY]	
PK	message_id (SERIAL)
FK	sender_id -> users
FK	receiver_id -> users
FK	copy_id -> book_copies
	subject (VARCHAR 200)
	body (TEXT), is_read (BOOL)

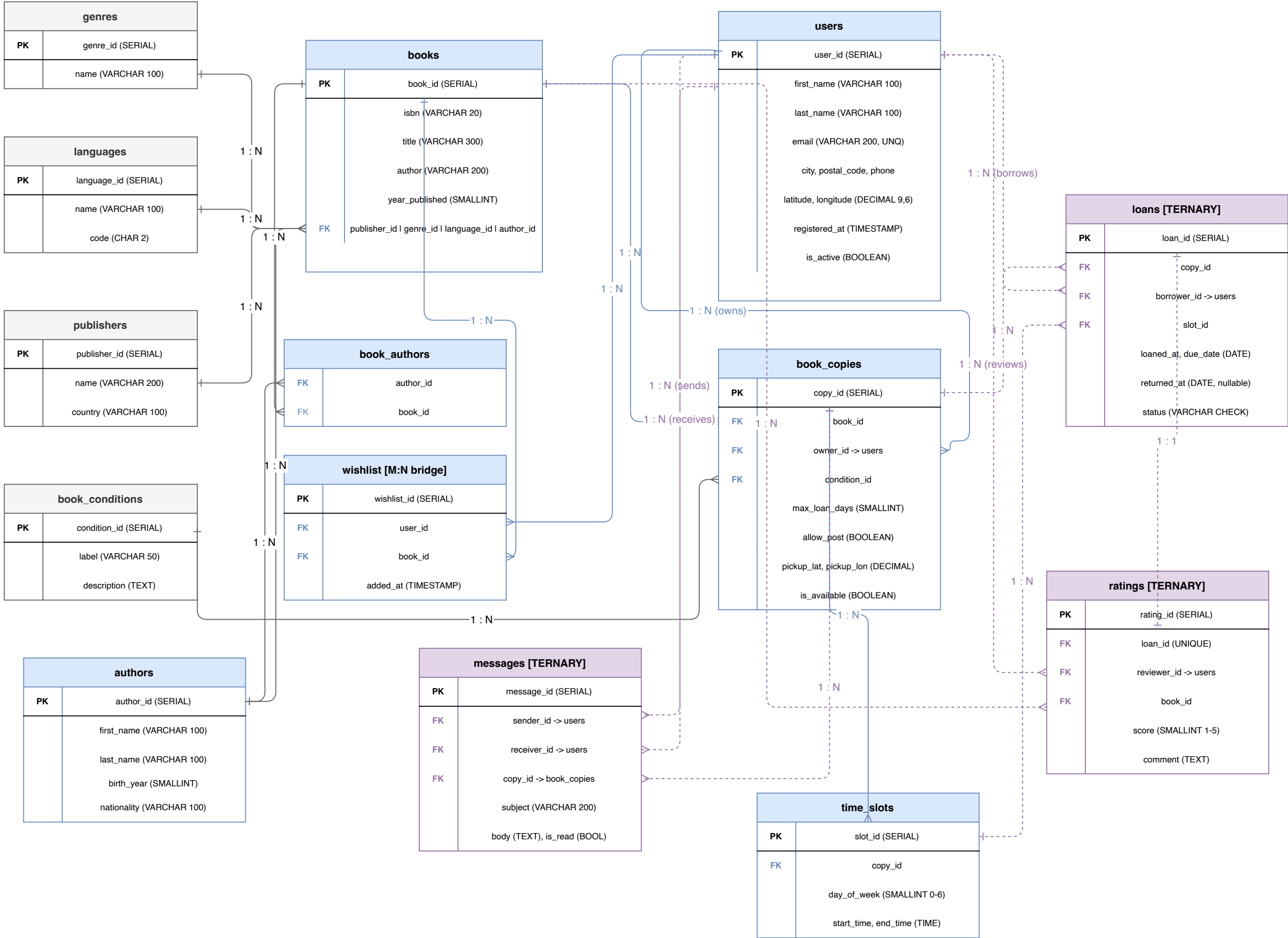
users	
PK	user_id (SERIAL)
	first_name (VARCHAR 100)
	last_name (VARCHAR 100)
	email (VARCHAR 200, UNQ)
	city, postal_code, phone
	latitude, longitude (DECIMAL 9,6)
	registered_at (TIMESTAMP)
	is_active (BOOLEAN)

book_copies	
PK	copy_id (SERIAL)
FK	book_id
FK	owner_id -> users
FK	condition_id
	max_loan_days (SMALLINT)
	allow_post (BOOLEAN)
	pickup_lat, pickup_lon (DECIMAL)
	is_available (BOOLEAN)

time_slots	
PK	slot_id (SERIAL)
FK	copy_id
	day_of_week (SMALLINT 0-6)
	start_time, end_time (TIME)

loans [TERNARY]	
PK	loan_id (SERIAL)
FK	copy_id
FK	borrower_id -> users
FK	slot_id
	loaned_at, due_date (DATE)
	returned_at (DATE, nullable)
	status (VARCHAR CHECK)

ratings [TERNARY]	
PK	rating_id (SERIAL)
FK	loan_id (UNIQUE)
FK	reviewer_id -> users
FK	book_id
	score (SMALLINT 1-5)
	comment (TEXT)



Datenwoerterbuch - Buchtausch-App

genres			
Attribut	Datentyp	Constraints	Beschreibung
genre_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
name	VARCHAR(100)	NOT NULL, UNIQUE	Genrebezeichnung

publishers			
Attribut	Datentyp	Constraints	Beschreibung
publisher_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
name	VARCHAR(200)	NOT NULL	Name des Verlags
country	VARCHAR(100)	optional	Land des Verlags

users			
Attribut	Datentyp	Constraints	Beschreibung
user_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
first_name	VARCHAR(100)	NOT NULL	Vorname
last_name	VARCHAR(100)	NOT NULL	Nachname
email	VARCHAR(200)	NOT NULL, UNIQUE	E-Mail (Login)
phone	VARCHAR(30)	optional	Telefonnummer
street	VARCHAR(200)	optional	Strasse und Hausnummer
city	VARCHAR(100)	optional	Stadt
postal_code	VARCHAR(20)	optional	Postleitzahl
country	VARCHAR(100)	DEFAULT Germany	Land
latitude	DECIMAL(9,6)	optional	GPS-Breitengrad
longitude	DECIMAL(9,6)	optional	GPS-Längengrad
registered_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Registrierungszeitpunkt
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Konto aktiv?

time_slots			
Attribut	Datentyp	Constraints	Beschreibung
slot_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
copy_id	INT	FK book_copies, NOT NULL	Verweis auf Exemplar
day_of_week	SMALLINT	NOT NULL, CHECK 0-6	Wochentag (0=Mo,6=So)
start_time	TIME	NOT NULL	Beginn Zeitfenster
end_time	TIME	NOT NULL, CHECK > start	Ende Zeitfenster

loans			
Attribut	Datentyp	Constraints	Beschreibung
loan_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
copy_id	INT	FK book_copies, NOT NULL	Ausgeliehenes Exemplar
borrower_id	INT	FK users, NOT NULL	Ausleiher
slot_id	INT	FK time_slots	Abholzeitslot
loaned_at	DATE	NOT NULL, DEFAULT TODAY	Startdatum
due_date	DATE	NOT NULL	Rückgabedatum
returned_at	DATE	optional	Tatsächl. Rückgabe
is_via_post	BOOLEAN	NOT NULL, DEFAULT FALSE	Per Post versendet?
status	VARCHAR(20)	CHECK active/returned/overdue	Status der Ausleihe

messages			
Attribut	Datentyp	Constraints	Beschreibung
message_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
sender_id	INT	FK users, NOT NULL	Absender
receiver_id	INT	FK users, NOT NULL	Empfänger
copy_id	INT	FK book_copies	Buchexemplar-Bezug
subject	VARCHAR(200)	optional	Betreff
body	TEXT	NOT NULL	Nachrichtentext
sent_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Sendezeitpunkt
is_read	BOOLEAN	NOT NULL, DEFAULT FALSE	Gelesen?

authors			
Attribut	Datentyp	Constraints	Beschreibung
author_id	INT	FK authors, NOT NULL	Verweis auf Autor
book_id	INT	FK books, NOT NULL	Verweis auf Buch

languages			
Attribut	Datentyp	Constraints	Beschreibung
language_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
name	VARCHAR(100)	NOT NULL, UNIQUE	Vollst. Sprachname
code	CHAR(2)	NOT NULL, UNIQUE	ISO-639-1-Code

book_conditions			
Attribut	Datentyp	Constraints	Beschreibung
condition_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
label	VARCHAR(50)	NOT NULL, UNIQUE	New/Good/Fair/Poor
description	TEXT	optional	Zustandsbeschreibung

books			
Attribut	Datentyp	Constraints	Beschreibung
book_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
isbn	VARCHAR(20)	UNIQUE	Standardbuchnummer
title	VARCHAR(300)	NOT NULL	Buchtitel
author	VARCHAR(200)	NOT NULL	Autor(in)
publisher_id	INT	FK publishers	Verweis auf Verlag
genre_id	INT	FK genres	Verweis auf Genre
language_id	INT	FK languages	Verweis auf Sprache
year_published	SMALLINT	optional	Erscheinungsjahr
description	TEXT	optional	Kurzbeschreibung
author_id	INT	FK authors, NOT NULL	Verweis auf Autor

book_copies			
Attribut	Datentyp	Constraints	Beschreibung
copy_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
book_id	INT	FK books, NOT NULL	Verweis auf Buchtitel
owner_id	INT	FK users, NOT NULL	Verweis auf Besitzer
condition_id	INT	FK book_conditions, NOT NULL	Zustand
max_loan_days	SMALLINT	NOT NULL, DEFAULT 14	Max. Ausleihdauer (Tage)
allow_post	BOOLEAN	NOT NULL, DEFAULT FALSE	Postversand?
pickup_address	VARCHAR(300)	optional	Abholadresse
pickup_latitude	DECIMAL(9,6)	optional	GPS-Breitengrad Abholort
pickup_longitude	DECIMAL(9,6)	optional	GPS-Längengrad Abholort
is_available	BOOLEAN	NOT NULL, DEFAULT TRUE	Ausleihbar?
added_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Einstellungszeitpunkt

ratings			
Attribut	Datentyp	Constraints	Beschreibung
rating_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
loan_id	INT	FK loans, NOT NULL, UNIQUE	Ausleihe (1:1)
reviewer_id	INT	FK users, NOT NULL	Bewertender Nutzer
book_id	INT	FK books, NOT NULL	Bewertetes Buch
score	SMALLINT	NOT NULL, CHECK 1-5	Bewertungspunkte
comment	TEXT	optional	Freitextkommentar
rated_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Bewertungszeitpunkt

wishlist			
Attribut	Datentyp	Constraints	Beschreibung
wishlist_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
user_id	INT	FK users, NOT NULL	Nutzer
book_id	INT	FK books, NOT NULL	Gewünschtes Buch
added_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Datum hinzugefügt

authors			
Attribut	Datentyp	Constraints	Beschreibung
author_id	SERIAL (INT)	PK, NOT NULL	Primärschlüssel
first_name	VARCHAR(100)	NOT NULL	Vorname des Autors
last_name	VARCHAR(100)	NOT NULL	Nachname des Autors
added_at	birth_year	optional	Geburtsjahr des Autors
nationality	VARCHAR(100)	optional	Nationalität des Autors